

Question:

Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How long does it take to download the file?

Reasoning + Answer:

<think>

The file size is 200 GB.

40% of 200 GB is:

$$\left[200 \times \frac{40}{100} = 80 \text{ GB} \right]$$

So, the download takes 80 minutes to reach 40% of the way through the download.

Since she has to restart from the beginning and the download speed is 2 GB/minute, the time to download the remaining 120 GB (since 200 - 80 = 120) is:

$$\left[120 \times \frac{1}{2} = 60 \text{ minutes} \right]$$

Thus, the total time to download the file is the time taken to download the first 80 GB plus the 20 minutes taken to restart, plus the time to download the remaining 120 GB:

$$\left[80 + 20 + 60 = 160 \text{ minutes} \right]$$

<answer>160</answer>

